AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

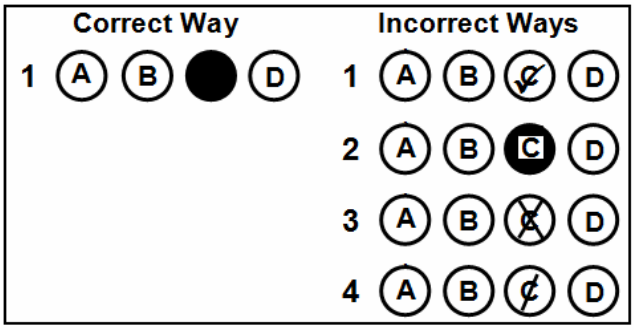
MAY 2014

Physics Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

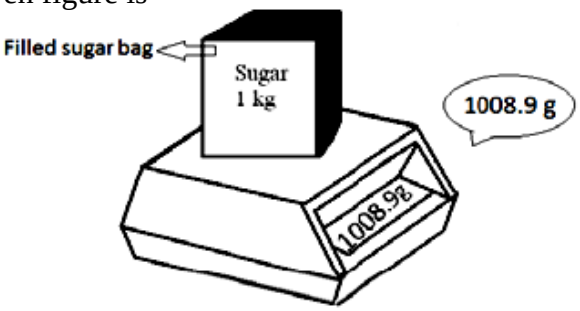
5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. Which of the following is the CORRECT number of significant figures and the standard form of a radius 0.0340 cm?

	Significant Figures	Standard Form
A	3	3.4×10^{-3} cm
B	3	3.40×10^{-2} cm
C	4	3.4×10^{-3} cm
D	4	3.4×10^{-2} cm

2. The mass of the empty sugar container in the given figure is

A. 8.9 g
B. 10 g
C. 100 g
D. 800 g



3. All of the following describe the role of physics in technology EXCEPT

A. the use of vaccine.
B. the use of internet.
C. the use of electronics.
D. the use of measuring instruments.

4. When an object of mass 2 kg is dropped from a 125 m high building, then it will reach the ground in (where g = 10 m/s^2)

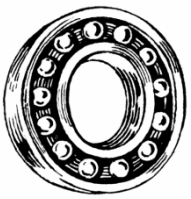
A. 5 s.
B. 10 s.
C. 20 s.
D. 40 s.

5. If a car travels at an average speed of 80 m/s in a race and covers a distance of 640,000 m, then the time taken by the car will be

A. 800 s
B. 8,000 s
C. 6,400,000 s
D. 51,200,000 s

6. Ball bearings are used in a variety of light and heavy machineries because they have small

A. size.
B. mass.
C. shape.
D. area in contact.



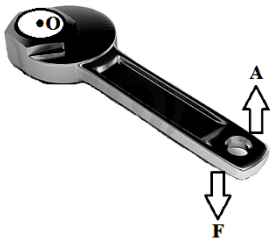
7. An object is moving with a uniform speed in a circle. If the radius of the path is doubled, then the centripetal force will

   A. reduce to half.
   B. become double.
   C. remain the same.
   D. reduce four times.

8. Two spheres **A** and **B** are dropped from the same height. If sphere **B** has a mass three times greater than sphere **A**, then which of the following statements CORRECTLY describes this situation?

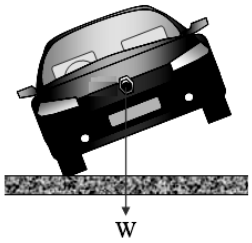
   A. Sphere **B** falls faster.
   B. Sphere **A** has more momentum.
   C. Sphere **B** has more momentum.
   D. Both spheres have the same momentum.

9. If a person is tightening a bolt by applying a force of 10 N at point 0.8 m from point **O**, then the turning effect of this force will be



   A. 8.0 Nm
   B. 9.2 Nm
   C. 10.8 Nm
   D. 12.5 Nm

10. When a car is tilted from the ground, then the center of gravity will

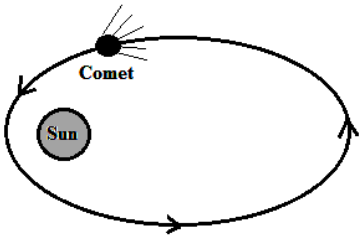


   A. remain the same.
   B. be lowered.
   C. be toppled.
   D. be raised.

11. Which of the following actions does NOT make use of the turning effect of a force?

   A. Tightening a tap
   B. Releasing a nut with a spanner
   C. Dabbing paint with a paint brush
   D. Picking up an ice cube with an ice tong

12. The given diagram shows the path followed by a comet when it reaches close to the sun. The shape of the path is



   A. elliptical.
   B. spherical.
   C. parabolic.
   D. hyperbolic.

13. If the mass of the earth is 6×10^{24} kg, its radius is 6.4×10^{6} m and the value of gravitational constant is 6.67×10^{-11} Nm^2 / kg^2, then the value of gravitational force acting on a 1.00 kg object when it is placed on surface of earth will be

A. 8.8 N
B. 9.8 N
C. 10.8 N
D. 11.8 N

14. The necessary acceleration for a satellite orbiting the earth is provided by

A. frictional force.
B. coulomb's force.
C. centrifugal force.
D. gravitational force.

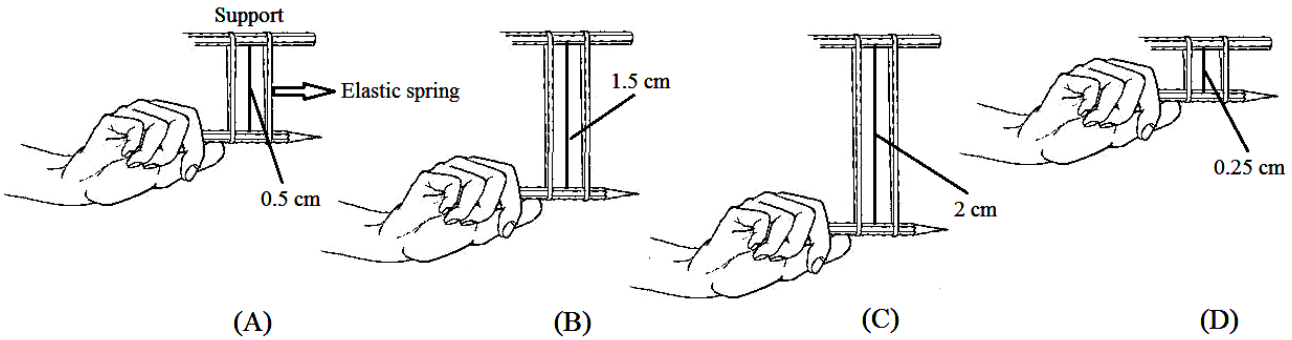
15. On which of the following factors does the pressure inside a liquid depend?

A. Force and area
B. Depth and force
C. Density and depth
D. Upthrust and density

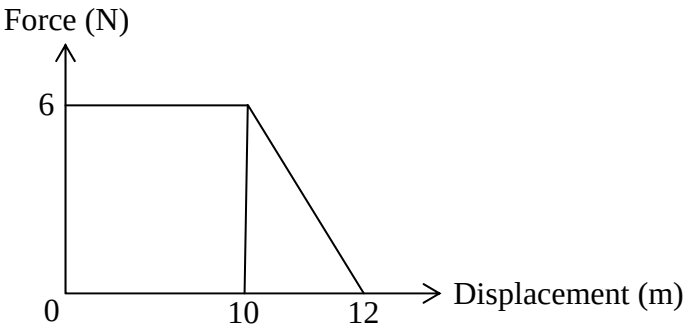
16. If a spring is stretched by hanging a piece of metal from it, then the energy stored in the spring will be

A. mechanical energy.
B. gravitational energy.
C. elastic potential energy.
D. chemical potential energy.

17. In which of the following figures, does the elastic spring have the greatest elastic potential energy?



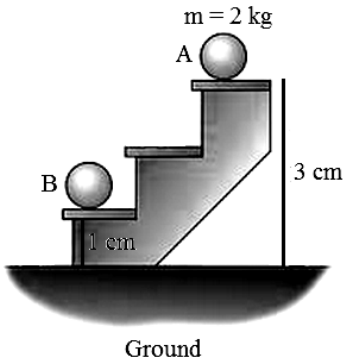
18. The given graph shows force **F** acting on a moving object.



What will be the work done on the body during its displacement of 12 m?

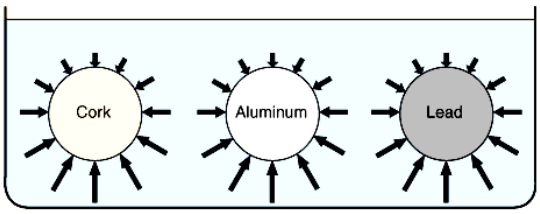
A. 36 J
B. 48 J
C. 66 J
D. 72 J

19. A ball of mass 2 kg is moving up from point **B** to point **A**. Its potential energy at point **A** with respect to the ground will be



A. 0.1 J
B. 0.2 J
C. 0.6 J
D. 0.8 J

20. Three balls of the same diameter are immersed in water completely. Which of the following is TRUE about the upthrust exerted by water on the balls?



A. All balls have the same upthrust.
B. Cork ball has maximum upthrust.
C. Lead ball has maximum upthrust.
D. Aluminum ball has maximum upthrust.

21. People feel cool when they do not dry themselves after swimming because water

    A. is colder than air.
    B. is a good conductor of heat.
    C. evaporates and causes cooling.
    D. insulates them from the warm air.

22. If the temperature of a chicken pizza is 75°C, then its temperature in Fahrenheit scale will be

    A. 23.89°F
    B. 73.69°F
    C. 167°F
    D. 348°F

23. In which direction will a bimetallic strip bend if the upper metal has a smaller and lower has a greater rate of thermal expansion?

    A. Upward
    B. Forward
    C. Backward
    D. Downward

24. When a saucepan is placed on a burning stove, heat is transferred to liquid inside the saucepan by

    A. radiation.
    B. convection.
    C. sublimation.
    D. evaporation.

25. Heat is transferred from one body to another body due to the difference in their

    A. latent heat.
    B. temperature.
    C. specific heat.
    D. nature of substance.

END OF PAPER

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